

FOC Motor Controller — Rev B Review Summary

Engineering issues, calculations to perform, and inverter design guidance

Target motor: ODrive M8325s 100KV (20 pole pairs, 92mm, 840g, pancake outrunner)

1. Issues With the Current Design

1.1 Critical — must resolve before layout

- DRV8323 part number as drawn does not match a real TI ordering code. For SPI control with no integrated buck, the correct part is **DRV8323SRTVR** (32-pin WQFN, RTV package). Update symbol value and footprint.
- CPH/CPL bootstrap capacitor voltage rating is insufficient. TI's datasheet rule is 2x margin over bus voltage; at 60V bus this requires a 120V+ rated cap. A 100V-rated part is undersized — use 200V or 250V rated 47nF X7R.
- DVDD decoupling capacitor is specified at 100nF; TI datasheet requires 1uF X5R/X7R, 6.3V, on DVDD to AGND. This is a stability requirement for the internal 3.3V core regulator, not optional.
- VM local bulk decoupling does not meet the datasheet minimum of 10uF X7R/X5R directly at the VM pin (CVM2 requirement). Add a 10-22uF 100V ceramic directly at the DRV8323 VM pin, separate from the MOSFET-stage bulk bank.
- MOSFET gate charge (Qg) vs DRV8323 charge pump capacity must be checked at your chosen switching frequency before finalizing IDRIVE/dead-time settings (see Section 2 for the calculation).

1.2 High priority

- Verify PA6 is assigned to TIM1_BKIN (AF6) in CubeMX, not SPI1_MISO (AF5), and that Break polarity is Active Low (nFAULT is open-drain active-low).
- STM32G431 pins PB4 and PB6 have an internal UCPD dead-battery pull-down (5.1k) enabled by default. Add `PWR_CR3 |= PWR_CR3_UCPD_DBDIS` in early firmware init if these pins are used for SPI1_MISO / DRV_CS.
- Confirm ADC1/ADC2 are configured in Dual Regular Simultaneous mode, triggered from TIM1_TRGO2 at the center-aligned PWM counter event, so ISOA and ISOB are sampled at the same instant.
- 1N4148 diodes in the asymmetric gate-drive path are marginal at 1A IDRIVE (near their peak current rating). Replace with BAT54S or PMEG2010 Schottky for margin and slightly faster turn-off.
- Snubber resistor power rating should be reverified once switching frequency is finalized (see Section 2, calculation 5). At higher frequencies 1/8W parts will exceed rating.
- LED series resistors (100R) push roughly 28mA through 0603 LEDs rated for less; change to 220R-330R.
- Battery sense clamp diode should be a zener (MM3Z3V3T1G) oriented cathode-to-signal, not a signal diode, for reliable ADC protection against transients.

1.3 Medium priority / platform completeness

- No reverse-polarity protection on the XT60 input. A P-channel ideal diode (e.g. SI7157DP) is cheap insurance for a board that will be plugged/unplugged often.
- No CAN node-ID selection hardware (DIP switch or solder jumpers) for a multi-board CAN network — currently ID must be set only in firmware.

- Confirm motor phase pads are laid out for the actual conductor size needed (see Section 3) since phase output is direct-solder pads rather than a connector.

1.4 Resolved / confirmed correct in this revision

- LMR38010FDDAR confirmed 80V input rated — no change needed for a 60V bus.
- AUX 7-pin JST-GH already added, wired to the same SPI bus as the onboard AS5047P — correct for a remote/auxiliary absolute encoder.
- Separate A3V3 analog rail from a dedicated LDO for ADC/VDDA is good practice and should be kept.
- Dual XT60 (M+F) for power daisy-chaining, CAN daisy-chain with ESD protection, and six fiducials are all production-appropriate choices — keep as is.

2. Engineering Calculations To Perform Before Layout

These are calculations you should do yourself with your actual chosen MOSFET's datasheet numbers. Formulas and reasoning are given; plug in your part's real Qg, Coss, Rds(on), and your firmware's target switching frequency.

2.1 Charge pump / gate drive capacity check

The DRV8323 charge pump must supply enough current to fully charge all six MOSFET gates every switching cycle. TI's design equation for sinusoidal (space-vector / FOC) commutation:

$$I_{VCP} > 3 \times Q_g \times f_{PWM}$$

Look up I_{VCP} from the DRV8323 datasheet curve of charge pump output current vs VM voltage (it decreases as bus voltage rises). Look up Q_g (total gate charge at your intended Vgs, usually 10V) from your MOSFET datasheet. Solve for the maximum safe f_{PWM} . If your desired switching frequency exceeds this, either lower f_{PWM} or choose a lower- Q_g MOSFET.

2.2 Gate resistor sizing

Pick a target turn-on time (100-300ns is a reasonable starting range per TI's own design example), then size gate resistance from:

$$t_{rise} \approx Q_{gd} / I_{gate} \quad I_{gate} = V_{drive} / (R_{gate_ext} + R_{gate_internal})$$

$R_{gate_internal}$ is in your MOSFET datasheet. Iterate: pick R_{gate} , compute I_{gate} at your chosen IDRIVE setting, compute expected rise time, compare against your dead-time budget.

2.3 Switching and conduction loss / thermal budget

For each MOSFET, at your worst-case phase current I and bus voltage V :

$$P_{conduction} = I_{rms}^2 \times R_{ds(on)} \times duty \quad P_{switching} = 0.5 \times V \times I \times (t_{rise} + t_{fall}) \times f_{PWM}$$
$$P_{total} = P_{conduction} + P_{switching} \quad (\text{per device})$$

Sum P_{total} across all 6 MOSFETs, add shunt resistor loss ($I^2 \times R_{shunt} \times 3$) and gate-drive loss ($Q_g \times V_{gs} \times f_{PWM} \times 3$). This is your total inverter-stage dissipation — use it in the thermal calculation below.

2.4 Thermal resistance path and junction temperature

Using your MOSFET's datasheet $R_{th(j-c)}$, and your intended mounting (bare PCB vs chassis-mounted with thermal pad):

$$T_j = T_{ambient} + P_{total_per_device} \times (R_{th_jc} + R_{th_interface} + R_{th_sink_ambient})$$

Compare T_j against the datasheet $T_j(max)$, leaving margin (design target usually $T_j \leq 125^\circ C$ even if the part is rated to $150-175^\circ C$). If bare-PCB dissipation gives an unrealistic T_j , this tells you a chassis/heatsink contact is mandatory, which then becomes a mechanical design requirement, not just electrical.

2.5 Snubber component sizing (RC snubber across switch node or Vds)

Estimate parasitic loop inductance from your layout (rule of thumb $\sim 1nH/mm$ of loop trace length, plus MOSFET package inductance from the datasheet). Then:

$$f_{ring} = 1 / (2 \times \pi \times \sqrt{L_{loop} \times C_{oss}}) \quad R_{snub} = \sqrt{L_{loop} / C_{oss}} \quad (\text{critical damping})$$
$$C_{snub} = 1x \text{ to } 3x \ C_{oss} \quad P_{snub} = C_{snub} \times V_{bus}^2 \times f_{PWM} \quad (\text{resistor power rating})$$

C_{oss} must be read from the datasheet capacitance-vs-Vds curve at your actual operating voltage, not the 25V test condition figure alone.

2.6 Miller-induced shoot-through margin

Check that the dV/dt during the complementary MOSFET's turn-on cannot pull the "off" MOSFET's gate above its threshold voltage through Cgd:

$$dV_{\text{induced}} = [C_{gd} / (C_{gd} + C_{gs})] \times (dV/dt) \times R_{\text{gate_off}}$$

Confirm dV_{induced} stays well below $V_{gs(th)}$. Use C_{gd}/C_{gs} values from the capacitance-vs- V_{ds} curve at your bus voltage, not the small-signal datasheet headline numbers.

2.7 Bulk capacitance sizing (MLCC DC-bias derating)

MLCC capacitance drops substantially under DC bias — often to 30-50% of nominal at your operating voltage. Pull the DC-bias derating curve from your specific capacitor's datasheet and compute effective capacitance, not nominal:

$$C_{\text{effective}} = \sum (C_{\text{nominal}_i} \times \text{derating_factor}_i(V_{\text{bias}}))$$

Target minimum effective bulk capacitance from ripple current and allowable bus ripple voltage:

$$C_{\text{min}} = I_{\text{ripple_peak}} / (f_{\text{PWM}} \times dV_{\text{allowed}})$$

Also check ripple current rating (A RMS) of your chosen bulk capacitor combination against your inverter's actual RMS ripple draw at worst-case modulation index.

2.8 Motor electrical parameter derivation (M8325s specific)

ODrive does not publish phase resistance/inductance for the M8325s. You will need to measure these yourself (or run the ODrive/your firmware's motor calibration routine) before finalizing current-loop gains:

- Phase resistance: measure line-to-line resistance with a precision milliohm meter or 4-wire measurement, divide by 2 for per-phase (wye) value.
- Phase inductance: measure with an LCR meter at your intended current-loop bandwidth frequency, or via a locked-rotor voltage-step test.
- Torque constant K_t : derive from KV rating: $K_t \text{ [Nm/A]} \approx 8.27 / \text{KV [RPM per V]}$. For 100KV: $K_t \approx 0.083 \text{ Nm/A}$ (verify empirically — this is a first-order estimate).
- Back-EMF constant relates directly to KV: at any operating speed, back-EMF must stay below available bus voltage headroom for correct commutation, particularly relevant if you push to high speeds on this pancake motor's 20 pole-pair count (high electrical frequency at moderate mechanical RPM).
- Electrical frequency at a given mechanical RPM: $f_{\text{elec}} = (\text{RPM}/60) \times \text{pole_pairs} = (\text{RPM}/60) \times 20$. This directly drives your minimum ADC sampling rate and current-loop bandwidth requirement, and should be cross-checked against your f_{PWM} choice from Section 2.1.

3. Inverter Stage Design Guidance (Theory and Resources)

This section is guidance and reference material only — the design decisions and final values are yours to work out.

3.1 MOSFET selection criteria (redo with your final part choice)

- Voltage margin: choose V_{ds} rating with at least 20-30% headroom over your worst-case bus transient, not just nominal bus voltage.
- $R_{ds(on)}$ vs Q_g tradeoff (Figure of Merit): a lower $R_{ds(on)}$ part generally has more die area and thus higher Q_g . For a 10A-class design you do not need the lowest $R_{ds(on)}$ parts on the market — those are optimized for 100A+ applications and carry Q_g that will exceed your gate driver's charge pump budget (see calculation 2.1). Look for parts balanced for your actual current range.
- Package thermal path: prefer packages with an exposed pad / bottom-side thermal pad suited to via-in-pad heat spreading into a copper pour and, ideally, a chassis contact.
- Reverse recovery charge (Q_{rr}) of the body diode matters if the body diode conducts during dead time — check this figure especially if your dead time is long relative to switching period.

3.2 Switching frequency selection — the central tradeoff

Your switching frequency choice cascades into nearly every other decision in this section: gate drive charge pump loading, switching losses, snubber sizing, filter cutoff, EMI, and audible motor noise.

Lower f_{PWM} (e.g. 15-20kHz)	Higher f_{PWM} (e.g. 30-40kHz)
Lower switching loss	Higher switching loss
Easier charge pump budget	Tighter charge pump budget (calc 2.1)
More audible motor noise	Ultrasonic, quieter
Coarser current ripple / torque ripple	Smoother current waveform
Simpler EMI filtering	More demanding EMI filtering

Given the M8325s has 20 pole pairs, your electrical frequency at speed will be high relative to a low pole-count motor — check calculation 2.8's f_{elec} formula against your target top speed and make sure f_{PWM} gives adequate switching cycles per electrical cycle (a common rule of thumb is at least 15-20x oversampling).

3.3 Gate drive network design theory

- Asymmetric turn-on/turn-off (series R with parallel diode) lets you slow down turn-on (reduce dV/dt induced shoot-through risk and ringing) while keeping turn-off fast (lower switching loss, faster fault response).
- IDrive and dead-time are programmable on the DRV8323S over SPI — this means you can tune switching behavior in firmware after board is built, rather than needing a resistor value change. Use this to your advantage: build test firmware that sweeps IDrive settings and observe gate waveforms before locking in a final value.
- Keep the physical loop area of driver-output → gate resistor → MOSFET gate → MOSFET source → shunt → driver-return as small as possible in layout. This loop is the primary radiator of high-frequency switching noise and also the source of parasitic inductance that causes ringing.

3.4 Snubber network theory

An RC snubber across the switching node (or drain-source of each FET) damps the LC ringing formed by parasitic loop inductance and the MOSFET's output capacitance (C_{oss}). The design goal is critical damping — enough resistance to kill the ring without excessive power dissipation. See calculation 2.5 for the sizing formulas. Two placement options exist: switch-node-to-ground (damps the whole half-bridge node) or drain-to-source across the

individual FET (damps that device specifically) — many designs use both; decide based on where your simulation or prototype testing shows the dominant ringing.

3.5 Current sense filter design theory

The RC filter between the shunt/DRV8323 current-sense amplifier output and the ADC input must: (a) pass the true current signal bandwidth with minimal phase delay, and (b) reject switching-frequency noise coupled from the adjacent switch node. Set the filter cutoff comfortably above your maximum expected current-loop bandwidth and f_{elec} (calc 2.8), but well below f_{PWM} for good attenuation of switching harmonics. Always sample the ADC at the center of the PWM period (center-aligned mode, triggered at counter peak/valley) so switching transients have settled before the sample is taken — filtering alone cannot fully substitute for correct sample timing.

3.6 Bulk capacitance and input filtering theory

The bulk capacitor bank's job is to source the high-frequency pulsed current the inverter draws that your battery/harness cannot supply fast enough through its own inductance. Undersizing this causes bus voltage sag/ripple during switching, which directly corrupts your voltage-reconstruction-based observer if you use one, and can also destabilize the gate driver's own internal regulators. See calculation 2.7 for sizing. A ferrite bead between the power connector and the bulk capacitor bank helps prevent switching noise from being conducted back up the battery harness into the rest of the robot's electronics.

3.7 Phase output pad sizing (direct-solder, no connector)

Since you're using direct-solder pads for phase output, size them using standard IPC-2221 current-carrying-capacity guidance for your expected continuous phase current, copper weight, and acceptable temperature rise. Also size the pad footprint for the wire gauge you intend to solder — for typical 10A-class phase current, 16-18 AWG silicone wire is a reasonable target, and the pad should be large enough to accept solder fillet plus strain relief (consider a small anchoring hole or wire-loop tie-down feature near the pad if the board will experience vibration in the leg).

3.8 Recommended reference material

- DRV8323 datasheet (SLVSDJ3D) Section 8/9 — external component table and detailed design procedure, including the charge pump and IDrive equations used above.
- Your MOSFET's datasheet — capacitance-vs- V_{ds} curves, gate charge curves, SOA and avalanche energy ratings, and thermal resistance/transient thermal impedance curves.
- TI application note SBAA698 (Understanding Gate Driver Slew Rate Control, MOSFET Switching Optimization and Protection Features) — covers pre-charge/pre-discharge gate drive concepts relevant to IDrive tuning.
- TI application report referenced in the DRV832x datasheet: "Architecture for Brushless-DC Gate Drive Systems" — background on the smart gate drive architecture.
- IPC-2221 — standard for PCB trace current-carrying capacity, used for phase pad and trace sizing.
- LTspice — build a half-bridge test circuit with your MOSFET's official SPICE model (usually available from the manufacturer's website) to validate gate drive, snubber, and shoot-through margin before committing to layout.

4. Things To Keep In Mind (User / System Level)

- The M8325s has no officially published current/torque rating — plan to determine your own safe continuous and peak current limits empirically (thermal testing on the actual motor) rather than assuming a figure.
- 20 pole pairs means high electrical frequency at even moderate mechanical speeds — verify your control loop and ADC sampling rate have adequate margin at your intended top speed (calc 2.8).
- This is a pancake / outrunner form factor — thermal path from the motor to ambient will differ from a cylindrical motor; if mounting the driver board directly to the motor back, consider thermal isolation or a dedicated heat spreader so motor heat does not add to the inverter's own thermal budget.
- Since phase output is direct-solder, plan your assembly/rework process now — reflow vs hand soldering heavy-gauge wire to a pad is a different process than connectorized assembly, and affects how easily the motor can be swapped during testing.
- Document every calculated value (Section 2) directly on the schematic as a text note next to the relevant component — this is what turns a set of component choices into a reviewable engineering design for a portfolio.
- Re-run the full ERC and a design review pass after implementing the critical/high-priority fixes in Section 1, before moving to layout.

End of summary.